

Domain Analysis & Spatial Evaluation Report

Jakob Balkovec & Kerry Cheon — Seattle University  
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|                                              |                                                 |                                          |                                               |
|----------------------------------------------|-------------------------------------------------|------------------------------------------|-----------------------------------------------|
| <div>31</div> <div>Monitoring Stations</div> | <div>63,165</div> <div>Total Observations</div> | <div>4</div> <div>Spatial Clusters</div> | <div>2017–2025</div> <div>Date Coverage</div> |
|----------------------------------------------|-------------------------------------------------|------------------------------------------|-----------------------------------------------|

# 1. Introduction

This report documents two sequential phases of our soil moisture modeling work: first, a domain analysis that groups monitoring stations into climate-coherent clusters; second, a spatial evaluation of MDR-TS v2.1, a Random Forest regressor trained to predict daily soil moisture at 5 cm depth. Together, these two phases answer a core question: does our model generalize beyond the handful of stations it was trained on, and if not, where exactly does it break down?

The dataset spans 31 weather and SCAN/USCRN monitoring stations distributed across Washington, Oregon, Idaho, and Montana, covering the period 2017 through 2025. All observations target the *soil\_moisture\_5cm* variable, measured in volumetric water content ( $\text{m}^3/\text{m}^3$ ). The combined train, validation, and test set totals 63,165 samples with no missing target values.

## 1.1 Objectives

- 1 Define dry, transition, and wet moisture regimes using data-driven quantile thresholds.
- 2 Cluster all 31 stations by regime composition and geography using a KNN approach.
- 3 Identify one representative domain for focused model training and evaluation.
- 4 Evaluate spatial generalization of MDR-TS v2.1 across in-domain and out-of-domain stations.
- 5 Pinpoint the stations and conditions where the model fails most severely.

# 2. Moisture Regime Definition

Before clustering stations, we needed a consistent, data-driven way to label each observation as belonging to a dry, transition, or wet regime. Rather than using fixed physical thresholds from the literature, we derived them empirically from the full dataset so that the three classes would be roughly balanced across all stations and time periods.

## 2.1 Quantile-Based Thresholds

Thresholds were set at the 33rd and 66th percentiles of the pooled *soil\_moisture\_5cm* distribution across all 63,165 observations. This approach deliberately avoids imposing external domain knowledge and keeps class sizes balanced, which matters for both the clustering step and any downstream regime-stratified evaluation.

| Regime     | Threshold     | Value ( $\text{m}^3/\text{m}^3$ ) | Sample Count | Share of Dataset |
|------------|---------------|-----------------------------------|--------------|------------------|
| Dry        | < 33rd pct    | < 0.0912                          | 20,845       | 33.0%            |
| Transition | 33rd–66th pct | 0.0912–0.2130                     | 20,790       | 32.9%            |

|     |                        |               |        |       |
|-----|------------------------|---------------|--------|-------|
| Wet | $\geq 66\text{th pct}$ | $\geq 0.2130$ | 21,530 | 34.1% |
|-----|------------------------|---------------|--------|-------|

Table 1. Quantile-based moisture regime thresholds and global class distribution.

Global statistics for the target variable: mean = 0.163, median = 0.151, standard deviation = 0.111, range = [0.000, 0.473]. The near-symmetric split confirms the thresholds are well-calibrated.

## 2.2 Statistical Validation

To confirm that the three regimes are genuinely separable rather than arbitrary labels, we ran two statistical tests. The Kolmogorov-Smirnov test on pairwise regime distributions returned a statistic of 1.000 in both comparisons (Dry vs. Transition and Transition vs. Wet), with p-values effectively at zero. This means the empirical distributions of the two groups share no overlap whatsoever once the threshold is applied.

Cohen's d, which measures effect size irrespective of sample count, was **d = 3.32** for Dry vs. Transition and **d = 3.16** for Transition vs. Wet. Values above 0.8 are conventionally labeled 'large'; values of this magnitude are rarely seen in real environmental datasets and give us strong confidence that the regime labels carry meaningful physical signal.

**Key finding:** The three moisture regimes are statistically well-separated (KS = 1.00, Cohen's d > 3.0 for both boundaries). The quantile thresholds at 0.0912 and 0.2130 m<sup>3</sup>/m<sup>3</sup> are appropriate and will remain fixed for all downstream analysis.

## 3. Station Clustering

With regime labels established, we characterized each station by three numbers: the fraction of its observations that fall into the Dry, Transition, and Wet regimes respectively. These percentages were combined with geographic coordinates (latitude and longitude) and standardized before clustering, so that both climate character and spatial proximity would influence which stations end up together.

### 3.1 Approach

We used a K-Nearest Neighbors approach anchored on five existing training stations: Darrington, Quinault, SourdoughGulch\_WA\_985, Spokane, and Touchet\_WA\_824. Rather than letting an unconstrained algorithm invent new cluster centers, each of the 26 remaining stations was assigned to whichever training station it was closest to in the five-dimensional feature space (Dry%, Transition%, Wet%, latitude, longitude, all standardized). This anchored clustering directly preserves the connection between cluster membership and the training data that generated the model.

Note that Touchet\_WA\_824 ended up assigned to the Quinault cluster rather than forming its own group. Its regime profile (56% Transition, 23% Wet) is a poor match for Quinault (57% Wet, 37% Transition) and likely reflects a genuine mismatch — Touchet behaves more like an interior transitional site than a coastal wet site. We flag this below.

### 3.2 Cluster Profiles

| Cluster               | Stations | Dry % | Trans % | Wet % | Mean SM |
|-----------------------|----------|-------|---------|-------|---------|
| Darrington            | 5        | 24.1% | 19.3%   | 56.7% | 0.225   |
| Quinault              | 4        | 8.7%  | 42.2%   | 49.1% | 0.215   |
| SourdoughGulch_WA_985 | 4        | 13.7% | 33.3%   | 53.0% | 0.212   |
| Spokane               | 18       | 45.0% | 33.6%   | 21.4% | 0.129   |

Table 2. Summary of the four spatial clusters. Mean SM = mean soil moisture ( $\text{m}^3/\text{m}^3$ ).

### 3.3 Cluster Descriptions

#### Darrington Cluster (5 stations)

The Darrington cluster groups stations in the wet, mountainous northwestern corner of Washington. On average, nearly 57% of observations are in the Wet regime and mean soil moisture sits at 0.225. Beyond Darrington itself, the cluster includes BeaverPass\_WA\_990, MartenRidge\_WA\_999, MFNooksack\_WA\_1011, and USCRN\_Darrington\_21\_NNE — all of which share the same high-moisture, low-variability character. This cluster is the natural setting for in-domain validation of a model trained primarily on Darrington data.

#### Quinault Cluster (4 stations)

Quinault represents a moderate-moisture environment with a notably high transition fraction (42%). The cluster includes USCRN\_Quinault\_4\_NE and USCRN\_Corvallis\_10\_SSW, both of which are coastal or near-coastal western Oregon stations with similar Wet-dominant profiles. The outlier here is Touchet\_WA\_824, assigned to this cluster by proximity despite having a substantially drier and more transitional character. Results from this cluster should be interpreted with that heterogeneity in mind.

#### SourdoughGulch Cluster (4 stations)

SourdoughGulch\_WA\_985 anchors a small cluster of four stations with wet-dominant profiles similar to Darrington but spanning a wider geographic range. The three OOD stations in this group — USCRN\_Dillon\_18\_WSW, SCAN\_ConradAgRc, and USCRN\_Coos\_Bay\_8\_SW — are geographically dispersed (southwestern Montana, north-central Montana, and coastal Oregon respectively), yet they share a high-moisture signal. The cluster is small and offers limited statistical power, but it confirms that the model's wet-biased behavior shows up consistently beyond the core training stations.

#### Spokane Cluster (18 stations)

Spokane is by far the largest and most heterogeneous cluster, pulling in 18 stations from eastern Washington, eastern Oregon, and the Snake River Plain of Idaho. Average Dry% is 45%, nearly double any other cluster, and mean soil moisture is 0.129 — well below the global mean of 0.163. This cluster represents the dry, semi-arid interior that the model was not optimized for. Several stations here — `BurntMountain_WA`, `SCAN_Lind_1`, and `SCAN_OrchardRangeSite` — are above 70% Dry, which is an extreme regime shift relative to the training distribution. The Spokane cluster is expected to be the primary failure zone in spatial evaluation.

**Domain choice:** For the purposes of in-domain evaluation, the **Darrington cluster** is the cleanest anchor — it has the highest regime coherence, a direct link to a training station, and a well-understood wet-dominated character. The **Spokane cluster** is the intended stress test, representing the largest domain shift from training conditions.

## 4. Spatial Model Evaluation — MDR-TS v2.1

MDR-TS v2.1 is a Random Forest regressor trained on 37 engineered features drawn from remote sensing data (Sentinel-1 SAR, Sentinel-2 optical, MODIS LST), SMAP soil moisture retrievals, precipitation and antecedent moisture indices, terrain variables, and seasonal encodings. The training set covers the period 2017–2020; the test set covers 2023–2025 and includes all 31 stations.

Spatial evaluation here means asking how the model performs on stations it has seen during training (the five original training stations) versus the 26 stations that are new to it. The design is intentional — the temporal split isolates time generalization, while the station-level comparison isolates spatial generalization.

### 4.1 Overall Performance

Across all 20,166 test observations from 31 stations, the model achieves an  $R^2$  of **-0.133**, a mean absolute error of **0.087 m<sup>3</sup>/m<sup>3</sup>**, and an RMSE of **0.120 m<sup>3</sup>/m<sup>3</sup>**. The negative  $R^2$  is a meaningful warning sign: it means the model explains less variance than a simple prediction of the global mean would. The positive bias of +0.039 m<sup>3</sup>/m<sup>3</sup> confirms a systematic over-prediction, consistent with a model that was trained predominantly on wetter stations.

That said, the overall number is heavily dragged down by a small number of stations with catastrophic errors. The story becomes much clearer when results are broken down by station and cluster.

### 4.2 Trained vs. Unseen Stations

| Group | n Stations | Mean $R^2$ | Mean MAE |
|-------|------------|------------|----------|
|-------|------------|------------|----------|

|                         |    |        |        |
|-------------------------|----|--------|--------|
| Trained stations (5)    | 5  | +0.615 | 0.0272 |
| New stations (25)       | 25 | −1.591 | 0.1188 |
| Gap (trained minus new) | —  | 2.206  | 0.0916 |

Table 3. Performance contrast between stations seen during training and unseen stations.

The gap is stark. On the five original training stations, the model is genuinely predictive (mean  $R^2 = 0.615$ , mean MAE = 0.027). On the 25 stations it has never encountered, performance collapses: mean  $R^2$  drops to -1.591 and mean MAE more than quadruples to 0.119. This is a textbook spatial generalization failure — the model has learned the behavior of a small set of locations well but has not extracted patterns that transfer to different landscapes.

### 4.3 Performance by Station

The table below lists per-station results for the full test set, sorted by  $R^2$ . Only six stations achieve a positive  $R^2$ , and only four of those are above 0.20.

| Station               | n     | $R^2$  | MAE   | Bias   |
|-----------------------|-------|--------|-------|--------|
| Spokane               | 897   | +0.940 | 0.022 | −0.003 |
| Darrington            | 999   | +0.890 | 0.025 | −0.017 |
| Quinault              | 1,044 | +0.759 | 0.027 | +0.021 |
| SourdoughGulch_WA_985 | 906   | +0.500 | 0.041 | −0.003 |
| USCRN_Arco_17_SW      | 707   | +0.258 | 0.057 | +0.004 |
| USCRN_Riley_10_WSW    | 849   | +0.226 | 0.042 | +0.026 |
| USCRN_John_Day_35_WNW | 932   | +0.171 | 0.020 | +0.011 |
| Touchet_WA_824        | 170   | −0.013 | 0.021 | +0.001 |
| USCRN_Spokane_17_SSW  | 890   | −0.147 | 0.099 | +0.070 |
| SCAN_TableMountain    | 552   | −0.191 | 0.053 | +0.025 |
| BeaverPass_WA_990     | 313   | −0.229 | 0.096 | −0.052 |
| BurntMountain_WA      | 491   | −0.320 | 0.068 | −0.040 |
| Paradise_WA           | 745   | −0.400 | 0.107 | +0.083 |
| USCRN_Murphy_10_W     | 836   | −0.474 | 0.055 | +0.055 |
| CayusePass_WA         | 764   | −0.517 | 0.135 | +0.108 |
| SCAN_Violet           | 776   | −0.579 | 0.074 | +0.072 |
| USCRN_St_Mary_1_SSW   | 424   | −0.784 | 0.086 | +0.073 |

|                         |       |        |       |        |
|-------------------------|-------|--------|-------|--------|
| MartenRidge_WA_999      | 505   | −1.053 | 0.139 | −0.109 |
| SCAN_OrchardRangeSite   | 922   | −1.113 | 0.046 | −0.043 |
| USCRN_Darrington_21_NNE | 806   | −1.200 | 0.114 | +0.108 |
| HartsPass_WA_515        | 239   | −2.015 | 0.209 | −0.177 |
| SCAN_Lind_1             | 1,004 | −2.000 | 0.057 | −0.056 |
| SCAN_CookFarmFieldD     | 511   | −2.097 | 0.166 | +0.166 |
| USCRN_Dillon_18_WSW     | 424   | −2.120 | 0.119 | +0.119 |
| RainyPass_WA_711        | 650   | −2.944 | 0.139 | −0.136 |
| SCAN_JordanValleyCwma   | 573   | −2.790 | 0.178 | +0.178 |
| SCAN_ConradAgRc         | 557   | −3.709 | 0.149 | +0.149 |
| USCRN_Quinault_4_NE     | 1,033 | −4.780 | 0.154 | +0.154 |
| USCRN_Corvallis_10_SSW  | 311   | −5.758 | 0.247 | +0.247 |
| USCRN_Coos_Bay_8_SW     | 331   | −6.797 | 0.242 | +0.242 |
| MFNooksack_WA_1011      | 5     | 0.000  | 0.238 | −0.238 |

Table 4. Per-station test set metrics, sorted by  $R^2$  (descending). Bias = mean signed error (positive = over-prediction).

## 5. Results by Cluster

Aggregating to the cluster level averages out station-level noise and reveals the underlying spatial generalization pattern more cleanly.

| Cluster               | Stations | Samples | $R^2$  | MAE   | Bias   |
|-----------------------|----------|---------|--------|-------|--------|
| Original (trained 5)  | 5        | 4,016   | +0.833 | 0.028 | ~0.000 |
| Darrington            | 5        | 2,628   | −0.244 | 0.083 | −0.001 |
| Spokane               | 18       | 12,762  | −0.082 | 0.079 | +0.026 |
| Quinault              | 4        | 2,558   | −2.182 | 0.105 | +0.101 |
| SourdoughGulch_WA_985 | 4        | 2,218   | −1.850 | 0.113 | +0.095 |

Table 5. Cluster-level aggregated test metrics.

### 5.1 The Original Five

Pooling the five training stations yields  $R^2 = 0.833$  and  $MAE = 0.028$  — strong performance with near-zero bias. These stations are genuinely well-modeled. The metric also establishes an upper-bound ceiling for what in-domain spatial generalization looks like.

## 5.2 Darrington Cluster

Despite being geographically close to the Darrington training station, the four OOD stations in this cluster are poorly predicted (cluster  $R^2 = -0.244$ ). The bias is essentially zero, which means the model is not systematically wrong in one direction — it is just imprecise. The wide Q5–Q95 error range (roughly  $\pm 0.21 \text{ m}^3/\text{m}^3$ ) suggests high spatial variability that the training station alone cannot capture, even within a climatically coherent region.

## 5.3 Spokane Cluster

The Spokane cluster — 18 stations, mostly dry interior sites — performs marginally better than Darrington in aggregate ( $R^2 = -0.082$ ), but this masks severe failures at individual stations. HartsPass\_WA\_515 ( $R^2 = -2.01$ ), RainyPass\_WA\_711 ( $R^2 = -2.94$ ), and the extreme dry-regime SCAN stations (SCAN\_Lind\_1 and SCAN\_OrchardRangeSite, both above 70% Dry) are practically unpredictable. The positive bias of  $+0.026$  across the cluster reflects the model systematically over-predicting soil moisture for dry sites, consistent with training data that is skewed toward wetter conditions.

## 5.4 Quinault and Sourdough Gulch Clusters

The two smaller wet-dominant clusters are counterintuitively the worst-performing groups overall ( $R^2 = -2.18$  and  $-1.85$  respectively). The culprits are specific stations: USCRN\_Quinault\_4\_NE ( $R^2 = -4.78$ ), USCRN\_Coos\_Bay\_8\_SW ( $R^2 = -6.80$ ), and USCRN\_Corvallis\_10\_SSW ( $R^2 = -5.76$ ). All three have a high Wet fraction and substantial positive bias (around  $+0.15$  to  $+0.25 \text{ m}^3/\text{m}^3$ ), meaning the model is severely over-predicting their already-high moisture values. One possible explanation is that these stations are in coastal or near-coastal environments that behave differently from the inland wet sites (Darrington, Quinault) the model learned from.

**Summary pattern:** Model quality degrades sharply the moment we move away from the original training stations. Cluster membership by regime similarity does *not* guarantee transferability — the Quinault-clustered coastal stations are among the worst performers despite appearing climatically similar to training data.

# 6. Failure Analysis

Across the evaluation, three distinct failure modes emerge. Understanding them separately is important because they likely require different fixes.

## 6.1 Systematic Over-Prediction at Coastal Wet Sites

USCRN\_Coos\_Bay\_8\_SW ( $R^2 = -6.80$ , bias = +0.242), USCRN\_Corvallis\_10\_SSW ( $R^2 = -5.76$ , bias = +0.247), and USCRN\_Quinalt\_4\_NE ( $R^2 = -4.78$ , bias = +0.154) all show enormous positive bias. The model is outputting soil moisture predictions that are too high by roughly 0.15–0.25 m<sup>3</sup>/m<sup>3</sup> on average. These sites are coastal Pacific Northwest locations with very different soil types, precipitation patterns, and drainage dynamics from the inland stations used for training. The model has apparently learned that high SAR/optical signals correlate with high soil moisture in inland settings, but this relationship breaks down near the coast where confounding factors (fog, marine-layer evapotranspiration, different soil textures) dominate.

## 6.2 Systematic Under-Prediction at Extreme Dry Sites

HartsPass\_WA\_515 (bias = -0.177), RainyPass\_WA\_711 (bias = -0.136), and MartenRidge\_WA\_999 (bias = -0.109) all show large negative bias — the model is under-predicting, outputting values that are too low by over 0.10 m<sup>3</sup>/m<sup>3</sup>. These are high-elevation, rain-shadow sites where soil moisture can drop very low during summer months. Since the training distribution has only 33% of samples in the Dry regime, the model has limited experience with the lower tail of the moisture distribution and tends to regress toward higher values.

## 6.3 High Variance Without Directional Bias

A third group — including BeaverPass\_WA\_990 ( $R^2 = -0.229$ , bias = -0.052) and USCRN\_Darrington\_21\_NNE ( $R^2 = -1.200$ , bias = +0.108) — shows poor  $R^2$  despite moderate absolute bias. The Q5–Q95 error ranges for these stations span roughly 0.40–0.45 m<sup>3</sup>/m<sup>3</sup>, indicating that prediction errors are large but not consistently in one direction. This is consistent with the model capturing a rough mean level but failing to track temporal dynamics — likely because the site-specific drivers (microclimate, soil depth, land cover) are not well-represented in the feature set.

## 6.4 Notable Exception: MFNooksack\_WA\_1011

MFNooksack\_WA\_1011 has only 5 test samples, making its  $R^2$  of 0.00 and bias of -0.238 statistically meaningless. This station should be excluded from any conclusions until more data is available.

# 7. Discussion and Recommended Next Steps

The clustering and spatial evaluation work together to paint a coherent picture. The model is well-suited to the interior Pacific Northwest landscapes it was trained on but degrades substantially when applied to stations with different regime compositions, coastal dynamics, or extreme dry conditions. The domain clustering helps us be precise about which stations fall outside the model's effective range and why.

## 7.1 Train on the Domain, Not the Full Dataset

The next logical experiment is to retrain MDR-TS using only the Darrington cluster as training data — a smaller, more coherent domain — and evaluate whether in-domain performance improves. The hypothesis is that removing the noisier, drier Spokane-cluster data from training will sharpen the model's ability to predict wet-regime conditions. This trades spatial breadth for spatial precision.

## 7.2 Regime-Stratified Training or Weighting

Given the observed over-prediction bias at wet coastal sites and under-prediction at dry sites, a promising direction is to either train separate models for each regime or apply sample weights that up-weight Dry-regime observations. Either approach forces the model to devote more capacity to the tails of the soil moisture distribution where errors are currently largest.

## 7.3 Investigate Feature Behavior at Failing Stations

For the three coastal stations with  $R^2$  below  $-5.0$ , it would be worth plotting feature distributions against the training set to understand what the model is actually receiving as input. If the feature distributions at Coos Bay or Corvallis are genuinely out-of-distribution relative to training, no amount of re-tuning will fix the predictions — the problem is input coverage, not model architecture.

## 7.4 Address the Touchet Misclassification

Touchet\_WA\_824 was assigned to the Quinault cluster by proximity in feature space, but its regime profile (56% Transition) is a clear mismatch with that cluster's wet character. It would be worth either re-running the clustering with Touchet excluded from the anchor set or reassigning it manually to the Spokane cluster, where its transitional character is better represented.

**Priority action:** Retrain on the Darrington domain and compare cluster-level  $R^2$  against the current v2.1 baseline. If in-domain  $R^2$  improves without catastrophic degradation elsewhere, domain-restricted training is a viable path forward. Run this before exploring architecture changes or additional feature engineering.

## Appendix A — Per-Station Regime Composition

The table below lists the full regime breakdown for all 31 stations, including their cluster assignment and mean soil moisture.

| Station                 | Cluster        | Dry % | Trans % | Wet % | Mean SM |
|-------------------------|----------------|-------|---------|-------|---------|
| BeaverPass_WA_990       | Darrington     | 20.3  | 15.1    | 64.6  | 0.237   |
| BurntMountain_WA        | Spokane        | 71.1  | 27.6    | 1.3   | 0.054   |
| CayusePass_WA           | Spokane        | 39.2  | 13.7    | 47.1  | 0.176   |
| Darrington              | Darrington     | 20.4  | 17.3    | 62.4  | 0.219   |
| HartsPass_WA_515        | Spokane        | 52.0  | 20.1    | 27.9  | 0.137   |
| MartenRidge_WA_999      | Darrington     | 34.4  | 20.7    | 45.0  | 0.203   |
| MFNooksack_WA_1011      | Darrington     | 15.5  | 17.6    | 66.9  | 0.286   |
| Paradise_WA             | Spokane        | 41.7  | 15.1    | 43.3  | 0.161   |
| Quinault                | Quinault       | 6.1   | 36.8    | 57.2  | 0.215   |
| RainyPass_WA_711        | Spokane        | 36.7  | 54.3    | 9.0   | 0.117   |
| SCAN_ConradAgRc         | SourdoughGulch | 3.5   | 42.2    | 54.2  | 0.223   |
| SCAN_CookFarmFieldD     | Spokane        | 25.8  | 28.1    | 46.0  | 0.204   |
| SCAN_JordanValleyCwma   | Spokane        | 34.7  | 33.8    | 31.5  | 0.158   |
| SCAN_Lind_1             | Spokane        | 72.4  | 26.9    | 0.7   | 0.057   |
| SCAN_OrchardRangeSite   | Spokane        | 70.2  | 27.2    | 2.6   | 0.067   |
| SCAN_TableMountain      | Spokane        | 25.2  | 57.3    | 17.5  | 0.149   |
| SCAN_Violett            | Spokane        | 25.3  | 46.8    | 27.9  | 0.166   |
| SourdoughGulch_WA_985   | SourdoughGulch | 12.1  | 23.3    | 64.6  | 0.239   |
| Spokane                 | Spokane        | 36.8  | 20.0    | 43.2  | 0.168   |
| Touchet_WA_824          | Quinault       | 20.9  | 55.9    | 23.2  | 0.159   |
| USCRN_Arco_17_SW        | Spokane        | 59.1  | 35.4    | 5.5   | 0.082   |
| USCRN_Coos_Bay_8_SW     | SourdoughGulch | 36.2  | 10.4    | 53.3  | 0.198   |
| USCRN_Corvallis_10_SSW  | Quinault       | 1.8   | 38.1    | 60.1  | 0.275   |
| USCRN_Darrington_21_NNE | Darrington     | 29.9  | 25.7    | 44.4  | 0.177   |

|                       |                |      |      |      |       |
|-----------------------|----------------|------|------|------|-------|
| USCRN_Dillon_18_WSW   | SourdoughGulch | 3.1  | 57.4 | 39.6 | 0.189 |
| USCRN_John_Day_35_WNW | Spokane        | 61.9 | 37.7 | 0.4  | 0.092 |
| USCRN_Murphy_10_W     | Spokane        | 28.4 | 56.1 | 15.5 | 0.139 |
| USCRN_Quinault_4_NE   | Quinault       | 6.2  | 38.1 | 55.7 | 0.212 |
| USCRN_Riley_10_WSW    | Spokane        | 57.3 | 38.4 | 4.4  | 0.095 |
| USCRN_Spokane_17_SSW  | Spokane        | 39.1 | 20.7 | 40.3 | 0.162 |
| USCRN_St_Mary_1_SSW   | Spokane        | 32.7 | 45.4 | 21.9 | 0.138 |

Table A1. Full station listing with cluster assignments and regime distributions.